Q1. Ethereum Consensus Mechanism

- Correct answer: Proof of Stake (PoS).
- Ethereum moved from PoW to PoS in 2022.
- PoS uses validators instead of miners.
- It consumes less energy.

Q2. Central Novelty of Bitcoin

- Correct answer: Proof of Work for decentralized consensus.
- PoW solved the double-spending problem.
- It allows trust without a central authority.

Q3. Properties of Money

- Medium of Exchange.
- Unit of Account.
- Store of Value.
- Security of Transfer.
- Resistance to Counterfeiting.

Q4. Cause of a Hard Fork

- Correct answer: Software change invalidating future blocks.
- Nodes disagree on rules.
- Blockchain splits into two chains.

Q5. Purpose of Gas in Ethereum

- Correct answer: To prevent infinite loops.
- Gas limits computation.
- Users pay for resources used.

Q6. Bitcoin Transactions

- Correct answer: Pseudonymous and traceable.
- Addresses are visible.
- Real identity is hidden but can be linked.

Q7. Ethereum Block Time

- Correct answer: Around 15 seconds.
- Faster than Bitcoin.
- Improves transaction speed.

Q8. Proof of Stake (PoS)

- PoS selects validators based on stake.
- Random selection avoids centralization.
- Validators lock coins as stake.
- Consumes much less power than PoW.
- Delegated PoS allows users to vote for validators.

Q9. Four Fundamental Concepts of Bitcoin

- Cryptography: Secures ownership and transactions.
- Proof of Work: Enables trustless consensus.
- Peer-to-Peer Network: Removes central authority.
- Distributed Ledger: Shared and transparent record.
- These concepts together enable decentralization.
- They were chosen because all are essential.
- Without any one, Bitcoin would not work securely.

Q10. Bitcoin Transaction Lifecycle

- User creates transaction.
- Transaction is signed with private key.
- Broadcasted to network.
- Miners verify transaction.
- Transaction added to a block.
- Block added to blockchain.
- Transaction gets confirmations.

Transaction Flow Diagram

User

↓

Broadcast to Network

↓

Miners Validate

↓

Block Creation

↓

Blockchain

Q11. Waiting for Confirmations

- Each confirmation adds security.
- Reduces risk of double spending.
- Makes transaction irreversible.
- Bitcoin usually requires 6 confirmations.

Q12. EU Crypto Law (MiCA – 2023)

- EU introduced MiCA regulation.
- Controls crypto mining and holding.
- Protects investors.
- Prevents money laundering.
- Requires crypto firms to register.

Q13. Initial Coin Offering (ICO)

- ICO is a fundraising method.
- Tokens sold to investors.
- Positive: Easy funding.
- Negative: Risk of scams.

Q14. Proof of Work Mining in Bitcoin

- Miners compete to solve hash puzzles.
- They try different nonce values.
- First valid hash wins.
- Block added to blockchain.
- Miner gets reward.
- Provides security but uses high energy.